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These solutions are not the only "perfect" or "10/10" solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Q1. On Homework #1 you proved that $(\mathbb{R}^+, \boxplus, \boxdot)$ was a vector space. Recall that,

$$\mathbb{R}^+ = \{x : x > 0\} \qquad x \boxplus y = xy \qquad c \boxdot x = x^c.$$

Prove that the function $T : \mathbb{R}^+ \to \mathbb{R}$ given by $T(x) = \ln(x)$ is a linear transformation.

Solution: Just note that for every $x, y, k \in \mathbb{R}$ we have that:

$$T(x \boxplus y) = \ln(xy) = \ln(x) + \ln(y) = T(x) + T(y)$$

and

$$T(k \boxdot x) = \ln(x^k) = k\ln(x) = kT(x).$$

Q2. In this question, you will investigate whether it is always possible to find a linear transformation

$$T : V \to V$$

such that $\text{image}(T) = \ker(T)$.

If it is possible, give an example. If it is impossible, explain why it is impossible.

(a) Does there exist $T : \mathbb{R}^4 \to \mathbb{R}^4$ such that $\text{image}(T) = \ker(T)$?

Solution: Consider the linear transformation given by

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_4 \\ x_3 \\ 0 \\ 0 \end{pmatrix}$$

Both the kernel and the image of T are equal to the set of vectors in $\mathbb{R}^4$ with 0s in their third and fourth coordinates.

(b) Does there exist $T : \mathbb{R}^5 \to \mathbb{R}^5$ such that $\text{image}(T) = \ker(T)$?

Solution: No. If $\text{image}(T) = \ker(T)$, then $\dim(\text{image}(T)) + \dim(\ker(T))$ is an even number, contradicting the Dimension Theorem.

(c) Suppose that V is finite dimensional. Prove a general statement about when there exists a linear transformation $T : V \to V$ such that image$(T) = \ker(T)$.

Solution:

- If V is odd-dimensional, then such a map cannot exist by the same argument used in (b).
- If V is even-dimensional, let $\{\mathbf{v}_1, \ldots, \mathbf{v}_{2n}\}$ be a basis for V, and consider the following linear transformation: If $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_{2n}\mathbf{v}_{2n}$, then $T(\mathbf{v}) = a_{2n}\mathbf{v}_1 + \ldots a_{n+1}\mathbf{v}_n$.

  In this case, image$(T) = \ker(T) = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.

In the next question, you will generalize the notion of image and ker.

Q3. Let V and W be vector spaces with $V' \subseteq V$ and $W' \subseteq W$ subspaces. Suppose that $T : V \to W$ is a linear transformation. Prove the following:

(a) $T(V') = \{T(\mathbf{v}') : \mathbf{v}' \in V'\} \subseteq W$ is a subspace of W.

Solution: Let $T(\mathbf{v}_1'), T(\mathbf{v}_2') \in T(V')$ and let $k \in \mathbb{R}$. Note that

$$T(\mathbf{v}_1') + kT(\mathbf{v}_2') = T(\mathbf{v}_1' + k\mathbf{v}_2') \in T(V')$$

given that V' is a subspace. Also, since $\mathbf{0}_V \in V'$, $\mathbf{0}_W = T(\mathbf{0}_V) \in T(V')$.
Thus, $T(V')$ is a subspace of W.

(b) $T^{-1}(W') = \{\mathbf{v}' : T(\mathbf{v}') \in W'\} \subseteq V$ is a subspace of V.

Solution: Let $\mathbf{v}_1', \mathbf{v}_2' \in T^{-1}(W')$ and let $k \in \mathbb{R}$. Since $T(\mathbf{v}_1'), T(\mathbf{v}_2') \in W'$, and W' is a subspace of W, we have that

$$T(\mathbf{v}_1' + k\mathbf{v}_2') = T(\mathbf{v}_1') + kT(\mathbf{v}_2') \in W',$$

so $\mathbf{v}_1' + k\mathbf{v}_2' \in T^{-1}(W')$. Also, $\mathbf{0}_V \in T^{-1}(W')$ given that $T(\mathbf{0}_V) = \mathbf{0}_W \in W'$.
Thus, $T^{-1}(W')$ is a subspace of V.

And now we consider the relationship between $T(V')$, $T^{-1}(W')$, $\ker(T)$, and image(T).

(a) For what subspace $W' \subseteq W$ does $T^{-1}(W') = \ker(T)$?

Solution: Note that

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\} = \{\mathbf{v} \in V : T(\mathbf{v}) \in \{\mathbf{0}\}\} = T^{-1}(\{\mathbf{0}\}).$$

(b) For what subspace $V' \subseteq V$ does $T(V') = \text{image}(T)$?

Solution: Note that

$$\text{image}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\} = T(V).$$

Q4. Consider the linear transformation: $D : \mathbf{P}_n(\mathbb{R}) \to \mathbf{P}_{n-1}(\mathbb{R})$ given by $D(p(x)) = \frac{d}{dx}[p(x)]$. Equivalently, $D(p(x)) = p'(x)$. Calculate the following.

(a) image (D)

Solution: The map D is surjective, i.e., image$(D) = \mathbf{P}_{n-1}(\mathbb{R})$: Given an arbitrary polynomial

$$p(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} \in \mathbf{P}_{n-1}(\mathbb{R}),$$

consider its anti-derivative

$$q(x) = a_0 x + \frac{a_1}{2}x^2 + \cdots + \frac{a_{n-1}}{n}x^n.$$

Then

$$\begin{aligned} D(q(x)) &= D\left(a_0 x + \frac{a_1}{2}x^2 + \cdots + \frac{a_{n-1}}{n}x^n\right) \\ &= a_0 D(x) + \frac{a_1}{2}D(x^2) + \cdots + \frac{a_{n-1}}{n}D(x^n) \\ &= a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} \\ &= p(x). \end{aligned}$$

(b) ker (D)

Solution: The kernel of D is the set of polynomials whose derivative is 0 everywhere, that is, the set of constant polynomials.

Q5. Let $T : V \to V$ be a linear transformation. A subspace $W \subseteq V$ is <u>T-invariant</u> if $T(W) \subseteq W$.
Prove the following subspaces are T-invariant.

(a) $\{\mathbf{0}\} \subseteq V$

Solution: Note that $T(\mathbf{0}) = \mathbf{0} \in \{\mathbf{0}\}$, so $T(\{\mathbf{0}\}) \subseteq \{\mathbf{0}\}$.

(b) V

Solution: For every $\mathbf{v} \in V$, we have that $T(\mathbf{v}) \in V$, so $T(V) \subseteq V$.

(c) image(T)

Solution: Let $\mathbf{v} \in$ image(T). Then $T(\mathbf{v}) \in$ image(T), and therefore $T(\text{image}(T)) \subseteq$ image(T).

(d) ker(T)

Solution: Let $\mathbf{v} \in \ker(T)$. Then $T(\mathbf{v}) = \mathbf{0} \in \ker(T)$, and therefore $T(\ker(T)) \subseteq \ker(T)$.

Q6. Let V be an n-dimensional vector space with basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. Suppose that $a_i \in \mathbb{R}$ for $i = 1, \ldots, n$. Define $T : V \to V$ by $T(\mathbf{v}_i) = a_i\mathbf{v}_i$.

(a) Compute $\ker(T)$.

Solution: We claim that $\ker(T) = \operatorname{span}\{\mathbf{v}_i : a_i = 0\}$:
($\subseteq$) Let $\mathbf{v} \in \ker(T)$ and let us express $\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$ so that

$$T(\mathbf{v}) = a_1b_1\mathbf{v}_1 + \cdots + a_nb_n\mathbf{v}_n = \mathbf{0}.$$

Since the $\mathbf{v}_i$'s are linearly independent, we have that $a_1b_1 = a_2b_2 = \cdots = a_nb_n = 0$. In particular, if $a_i \neq 0$, then $b_i = 0$, and therefore $\mathbf{v} \in \operatorname{span}\{\mathbf{v}_i : a_i = 0\}$.
($\supseteq$) If $\mathbf{v} \in V$ is a linear combination of only $\mathbf{v}_i$'s satisfying that $a_i = 0$, then

$$T(\mathbf{v}) = 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_n = \mathbf{0},$$

and therefore $\mathbf{v} \in \ker(T)$.

(b) Compute $\operatorname{image}(T)$.

Solution: We claim that $\operatorname{image}(T) = \operatorname{span}\{\mathbf{v}_i : a_i \neq 0\}$:
($\subseteq$) If $\mathbf{v} \in \operatorname{image}(T)$, let $\mathbf{u} \in V$ be such that $\mathbf{v} = T(\mathbf{u})$, and let us express

$$\mathbf{u} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$$

so that $\mathbf{v} = T(\mathbf{u}) = a_1b_1\mathbf{v}_1 + \cdots + a_nb_n\mathbf{v}_n$. From this equation, we can conclude that $\mathbf{v}$ is a linear combination of only $\mathbf{v}_i$'s for which $a_i \neq 0$.
($\supseteq$) If $\mathbf{v} \in \operatorname{span}\{\mathbf{v}_i : a_i \neq 0\}$, let us express

$$\mathbf{v} = \sum_{a_i \neq 0} b_i\mathbf{v}_i$$

and consider the vector

$$\mathbf{u} = \sum_{a_i \neq 0} \frac{b_i}{a_i}\mathbf{v}_i.$$

Then

$$T(\mathbf{u}) = T\left(\sum_{a_i \neq 0} \frac{b_i}{a_i}\mathbf{v}_i\right) = \sum_{a_i \neq 0} b_i\mathbf{v}_i = \mathbf{v},$$

and therefore $\mathbf{v} \in \operatorname{image}(T)$.